

Evaluating Hitter Sustainability Using Process Metrics

Executive Summary

This report explains why process statistics like zone contact rate, hard-hit rate, and swing decisions are better predictors than traditional outcomes. I show how to use SQL to get these metrics from Trackman or Hawkeye data and discuss how teams can use them for player evaluation, development planning, and trade decisions.

Why Do Process Stats Matter?

The Problem With Process Metrics

Traditional statistics cannot tell the whole story, especially in small sample sizes, and often fail to capture a hitter's underlying skills. Stats such as batting average, home runs, and OPS are heavily influenced by other contexts such as luck, which increases evaluation risk when it's the only thing used.

The Advantage of Process Metrics

In contrast, process statistics stabilize more quickly and provide clearer insight into a hitter's swing decisions, bat-to-ball skills, and quality of contact. These metrics are particularly valuable in amateur and lower minor league settings; scouts must project performance against higher competition levels.

The Difference Between the Two

While traditional statistics still provide descriptive value, process statistics offer better insight for projecting growth, assessing risk and variance, distinguishing luck from skill, and providing actionable feedback for player development.

Which Underlying Metrics I Would Prioritize

Contact Quality

The first process metric I would prioritize for a hitter is contact quality. Metrics **like hard-hit rate, average exit velocity, and barrel rate** are great indicators of whether a hitter's success is sustainable or just short-term success. **Strong contact quality** tells us how **efficient and repeatable a hitter's swing is**, even if the hits don't come. If a player hits the ball hard consistently, it gives evaluators confidence that production will follow over a **larger sample**.

Swing Decisions

The next area I would focus on is swing decision-making. **Chase rate and zone swing rate** help show if a hitter can avoid pitchers pitches while still being aggressive on strikes in the zone.

These metrics give insight into a **hitter's approach, strike-zone awareness, and how well that approach will be against higher level pitching**. Hitters who make good swing decisions have the potential for a **higher floor** because they see more pitches, miss less often, and avoid becoming **one-dimensional hitters**.

Bat-to-Ball Skills

Bat-to-ball skills are also an important part of the evaluation. Metrics such as **zone contact rate** help establish whether a hitter has the **foundation to put the bat on the ball**. These skills **stabilize quickly** and are often a part of **sustained** offensive success. Strong bat-to-ball ability can also point evaluators toward **mechanical strengths**, such as advanced bat speed, sequencing, and overall swing efficiency.

Batted-Ball Profile

Finally, batted-ball profile metrics give evaluators a clue to a hitter's potential offensive profile. **Fly-ball and ground-ball rates** can suggest whether a hitter is more likely to be a power or more contact-oriented profile. These rates **become serviceable faster than other metrics like BABIP** and they help show evaluators what kind of value a hitter can potentially bring to an organization.

Stabilization Rates

Hard Hit Rate	29 Games
Chase Rate	40-50 PA
Zone Contact Rate	100-150 PA
Fly Ball Rate	80-100 Balls in Play

Below is a comparison of process metrics versus traditional outcome stats.

Metric Comparison

Metric	Stabilization	Predictive Value	Teachability
Hard Hit Rate	29 Games	Very High	Low
Chase Rate	40-50 PA	High	High
Zone Contact	100-150 PA	High	Medium
BABIP	Full Season	Low	N/A

How The Data Would Be Queried Using SQL

Data Assumptions:

- Consistent Player Identifiers
- Hawkeye / Trackman-style pitch-level CSV structure
- Basic Pitch Context Available
- Batted-Ball Information Included
- Single Table Structure
- Small Sample Size

1. Contact Quality: How to Find a Hitter's Hard-Hit Percentage

```
SELECT batter,
       SUM(CASE WHEN PitchCall = 'InPlay' AND ExitSpeed >= 95 THEN 1 ELSE 0 END) AS HardHitBalls,
       SUM(CASE WHEN PitchCall = 'InPlay' THEN 1 ELSE 0 END) AS TotalBallsInPlay,
       ROUND(SUM(CASE WHEN PitchCall = 'InPlay' AND ExitSpeed >= 95 THEN 1 ELSE 0 END) * 100.0 /
             NULLIF(SUM(CASE WHEN PitchCall = 'InPlay' THEN 1 ELSE 0 END), 0), 2) AS HardHitRate
FROM your_table_name
GROUP BY batter
ORDER BY HardHitRate DESC;
```

2. Swing Decisions: How to Find In-Zone Swing Percentage

```
SELECT batter,
       COUNT(*) AS TotalPitchesInZone,
       SUM(CASE WHEN PitchCall IN ('StrikeSwinging', 'FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END) AS TotalSwings,
       ROUND(SUM(CASE WHEN PitchCall IN ('StrikeSwinging', 'FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS ZoneSwingPercentage
FROM your_table_name
WHERE PlateLocSide BETWEEN -0.85 AND 0.85
AND PlateLocHeight BETWEEN 1.5 AND 3.5
AND PitchCall IS NOT NULL
GROUP BY batter
ORDER BY ZoneSwingPercentage DESC;
```

3. Bat-To-Ball Ability: How to Find Zone Contact Percentage

```
SELECT
  batter,
  SUM(CASE WHEN PitchCall IN ('StrikeSwinging', 'FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END) AS ZoneSwings,
  SUM(CASE WHEN PitchCall IN ('FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END) AS ZoneContact,
  ROUND(
    SUM(CASE WHEN PitchCall IN ('FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END) * 100.0 /
    NULLIF(SUM(CASE WHEN PitchCall IN ('StrikeSwinging', 'FoulBallNotFieldable', 'InPlay') THEN 1 ELSE 0 END), 0),
    2
  ) AS ZoneContactRate
FROM your_table_name
WHERE PlateLocSide BETWEEN -0.85 AND 0.85
AND PlateLocHeight BETWEEN 1.5 AND 3.5
GROUP BY batter
ORDER BY ZoneContactRate DESC;
```

4. Batted Ball Profile: How to Find Groundball/Flyball Rate

```
SELECT
    batter,
    SUM(CASE WHEN TaggedHitType = 'GroundBall' THEN 1 ELSE 0 END) AS GroundBalls,
    SUM(CASE WHEN TaggedHitType = 'FlyBall' THEN 1 ELSE 0 END) AS FlyBalls,
    SUM(CASE WHEN TaggedHitType = 'LineDrive' THEN 1 ELSE 0 END) AS LineDrives,
    SUM(CASE WHEN TaggedHitType = 'Popup' THEN 1 ELSE 0 END) AS Popups,
    COUNT(TaggedHitType) AS TotalBBE,
    ROUND(SUM(CASE WHEN TaggedHitType = 'GroundBall' THEN 1 ELSE 0 END) * 100.0 /
        NULLIF(COUNT(TaggedHitType), 0), 1) AS GroundBallRate,
    ROUND(SUM(CASE WHEN TaggedHitType = 'FlyBall' THEN 1 ELSE 0 END) * 100.0 /
        NULLIF(COUNT(TaggedHitType), 0), 1) AS FlyBallRate
FROM your_table_name
WHERE PitchCall = 'InPlay'
    AND TaggedHitType IS NOT NULL
GROUP BY batter
ORDER BY FlyBallRate DESC;
```

How This Would Be Used in a Real Organization

- **Identify Breakout Candidates**
 - Identify Underperforming Hitters with High Expected Stats
- **Unsustainable Stats**
 - Flag Players with Unsustainable Stats or Overperformers
- **Support Promotion Decisions**
 - Support Promotion or Demotion Decisions in Small Samples
- **Player Development Plan**
 - Inform Individualized Player Development Plans
- **Lineup Versatility**
 - Assist with Lineup Construction and Platoon Usage

Limitations and Context

-While process metrics provide predictive value, they must be interpreted with context

- Must be Paired with Video and In-Person Evaluation
- Process Stats Help, but do not Overshadow Scouting
- Certain Metrics Stabilize Faster than Others
- Pitching and Competition Level Matter
- Small Samples can Impact Process Stats